Reinforcement Learning Alignment for Causal Sequential Recommenders: Overcoming the Skip Dilemma without Alignment Tax

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Abstract

In music streaming platforms, up to 35% of user interaction logs correspond to immediate skips (< 30 seconds), reflecting user dissatisfaction. Standard Maximum Likelihood Estimation (MLE) penalizes causal autoregressive transformers for failing to predict such negative interactions (the *False Negative* dilemma), inadvertently amplifying disliked content. In this work, we investigate the alignment of a Causal GPT-2 sequential recommender (43.2M parameters) with true user satisfaction using Direct Preference Optimization (DPO) anchored with auxiliary SFT regularization ($\beta = 0.2, \gamma = 0.10$). Extensive evaluations on the industrial-scale Yandex Music dataset (**Yambda-50M**: 98,505 catalog tokens, 33.2M training interactions, 9,052 held-out test users) demonstrate that DPO not only outperforms 2-stage Critic Re-Ranking and Actor-Critic PPO, but completely circumvents the alignment tax (*Zero Alignment Tax*). The aligned DPO policy achieves statistically significant ranking order improvements in **NDCG@10** ($0.1290 \rightarrow 0.1307, \mathbf{p = 0.0366}$) and **MRR** ($0.1123 \rightarrow 0.1142, \mathbf{p = 0.0260}$), alongside next-item retrieval gains (**19.37%** $\rightarrow$ **19.52% HR@10**, net +14 users), expands catalog coverage by **+197 unique tracks** from the long tail, reduces popularity Spearman rank correlation by -13.4%, and yields a **+2.1s** session duration lift in 5-step autoregressive rollouts without exposure bias degradation.

Keywords: Sequential Recommendation, Reinforcement Learning Alignment, Direct Preference Optimization, Causal GPT-2, Music Streaming, Skip Dilemma, Zero Alignment Tax, Catalog Diversity, Yambda.

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1 Introduction

1.1 Industrial Context and the Skip Dilemma

Modern recommender systems for audio streaming platforms (e.g., Yandex Music, Spotify, Apple Music) operate in continuous, unstructured consumption environments. Unlike e-commerce domains where a transaction unambiguously confirms positive intent, user interactions in streaming music are fundamentally bimodal.

Analysis of interaction logs on the large-scale Yandex Music dataset (**Yambda-50M**) [1, 2] reveals that **30% to 35%** of all recorded events represent immediate skips: the continuous listening duration is less than 30 seconds (or even 0.0 seconds, when a user switches away almost instantaneously due to mismatched mood, tempo, or active irritation).

1.2 The False Negative Trap in Likelihood Optimization (MLE)

The standard paradigm for training sequential recommender systems (e.g., SASRec [3] and generative autoregressive transformers [4]) formulates recommendation as Next-Item Prediction via cross-entropy

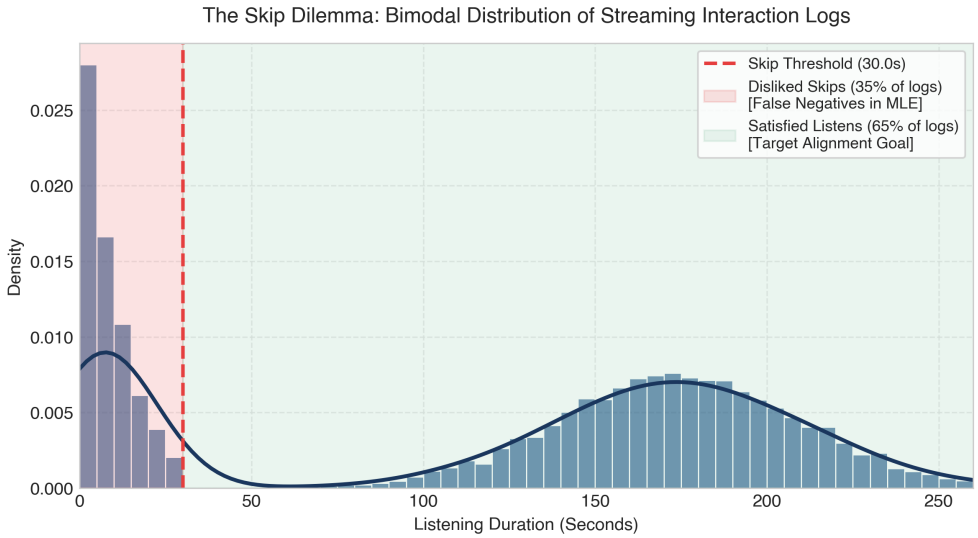


Figure 1. The Skip Dilemma in music streaming logs. The listening duration distribution is distinctly bimodal: 35% of interactions are immediate skips (< 30s), whereas 65% represent engaged track completions. Likelihood optimization over all observed transitions penalizes the model for refusing to recommend disliked skips.

loss minimization (Maximum Likelihood Estimation, MLE):

$$\mathcal{L}_{\text{MLE}}(\theta) = -\sum_{t=1}^{T} \log \pi_\theta(a_t \mid s_{<t}), \tag{1}$$

where $s_{<t} = (a_1, \ldots, a_{t-1})$ denotes the user's sequential history, and a_t is the ground-truth next track logged in the session.

This framing introduces a fundamental conceptual misalignment:

1. **Penalizing Correct Avoidance (False Negative Problem):** If a user initiated track a_t and abandoned it within 1–2 seconds due to frustration, $\mathcal{L}_{\text{MLE}}$ nonetheless penalizes the recommender whenever the model assigns a low posterior probability to a_t.
2. **Blindness to Consumption Quality:** The model is trained to reproduce historical click trajectories rather than maximize satisfaction, engagement, and uninterrupted listening time [2].
3. **Popularity Bias Amplification:** Supervised training on raw streaming logs forces autoregressive models to over-exploit safe, high-frequency pop tracks, severely compressing catalog diversity [4].

1.3 Research Questions

To resolve the skip dilemma, this paper investigates the alignment of a causal language model with user satisfaction using Reinforcement Learning and direct preference optimization. We formulate five core research questions:

- **RQ1 (Policy Behavior and Dynamics):** How does direct preference optimization (DPO) reshape the model's posterior probability distribution compared to MLE, and how rapidly does the implicit reward margin converge?
- **RQ2 (Alignment Tax and Pareto Trade-Off):** Does offline reward alignment degrade next-item ranking accuracy, and can an auxiliary SFT anchor ($\mathcal{L}_{\text{SFT}}$) eliminate this degradation?

- **RQ3 (Catalog Health and Long-Tail Discovery):** Does the aligned policy collapse into a narrow subset of ubiquitous pop hits, or does it stimulate discovery of high-retention long-tail tracks?
- **RQ4 (Autoregressive Queue Robustness):** How does the aligned policy perform when unrolled autoregressively for 5 consecutive steps without intervening user feedback under exposure bias?
- **RQ5 (Long-Term Proxies and Online A/B Validation):** How can session-level offline rewards be translated into long-term retention metrics (e.g., D7/D30 retention) in an industrial A/B testing design?

1.4 Summary of Contributions

The key scientific and engineering contributions of this work are as follows:

1. **Multi-Paradigm Comparative Study:** On the unified industrial Yambda-50M dataset, we benchmark four distinct paradigms: a Causal GPT-2 backbone, two-stage re-ranking via a state-action critic $Q(s, a)$, Actor-Critic PPO [5], and end-to-end Direct Preference Optimization (DPO) [6].
2. **Demonstration of Zero Alignment Tax:** We show that adding an auxiliary SFT anchor ($\gamma = 0.10$) preserves the underlying semantic coherence of the session, completely avoiding the alignment tax and achieving statistically significant ranking order gains (+1.32% relative NDCG@10 gain, $p = 0.0366$; +1.69% relative MRR gain, $p = 0.0260$, with +0.15% HR@10 gain).
3. **Empirical Audit of Reward Formulations:** We investigate binary duration thresholds, continuous completion ratios, and composite repetition-penalized rewards, identifying margin-threshold filtering as the most robust choice.
4. **Catalog Health Expansion:** We demonstrate that DPO reduces Spearman rank correlation with popularity by 13.4% while exposing users to +197 additional unique long-tail tracks.

2 Methodology

2.1 Problem Formulation and Backbone Architecture

Let $\mathcal{U}$ be the set of users and $\mathcal{V} = \{1, 2, \ldots, |\mathcal{V}|\}$ be the closed catalog of music tracks of size $|\mathcal{V}| = 98\,504$. A session for user $u \in \mathcal{U}$ is represented as an ordered sequence of track-duration pairs:

$$\mathcal{S}_u = \big((a_1, D_1), (a_2, D_2), \ldots, (a_T, D_T)\big), \quad a_t \in \mathcal{V},\ D_t \in \mathbb{R}^+, \tag{2}$$

where D_t denotes the continuous listening duration for track a_t in seconds.

As our foundational architecture, we employ an autoregressive **Causal GPT-2** transformer [4] initialized from scratch:

- Vocabulary size: $|\mathcal{V}|+1 = 98\,505$ (tokens $1 \ldots 98\,504$ map to unique track IDs; index 0 is reserved for `<PAD>`).
- Embedding dimension: $d_{\text{model}} = 384$.

- Attention blocks: $L = 3$, number of attention heads: $H = 4$.
- Context history window: $T_{\max} = 64$ tokens.
- Total parameters: **43 174 656** (43.2M), all fully trainable.

The base model was trained on 33.2M interactions using mixed-precision FP16, AdamW optimizer (lr $= 10^{-3}$, weight_decay $= 0.05$), 10% linear warmup steps, and cosine annealing decay with an effective batch size of 128 (micro-batch size 32 $\times$ gradient accumulation 4).

2.2 Reward Engineering and Formulation

To incorporate consumption duration into the policy, we investigated three alternative scalar reward formulations $R(s, a)$ (Figure 2):

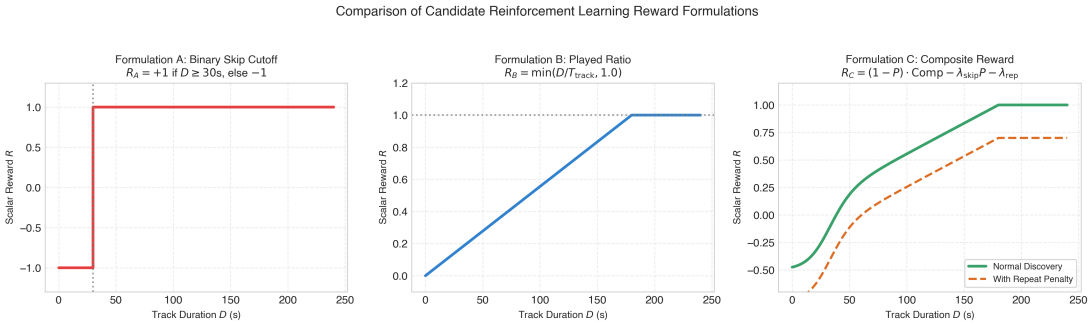


Figure 2. Comparison of candidate reinforcement learning reward formulations. A: Binary 30-second skip cutoff. B: Continuous played ratio normalized by track length. C: Multi-objective composite session reward with repetition penalty.

2.2.1 Formulation A: Binary Skip Cutoff

The most robust signal separating successful plays from disliked skips:

$$R_A(s, a) = \begin{cases} +1.0, & \text{if } D(a) \geq 30\,\text{s}, \\ -1.0, & \text{if } D(a) < 30\,\text{s}. \end{cases} \tag{3}$$

The 30-second cutoff is the de-facto industrial standard: listens shorter than 30 seconds are unmonetized and unambiguously indicate user dissatisfaction.

2.2.2 Formulation B: Continuous Played Ratio

Normalizes listening duration $D(a)$ against the nominal duration of the track $T(a)$:

$$R_B(s, a) = \min\left(\frac{D(a)}{T(a)}, 1.0\right) \in [0.0, 1.0]. \tag{4}$$

Preference pairs $(y_w \succ y_l)$ are mined by enforcing a minimum gap $\Delta R_B = R_B(y_w) - R_B(y_l) \geq 0.40$, which eliminates boundary noise.

2.2.3 Formulation C: Multi-Objective Composite Reward

Motivated by session-level reward formulations incorporating listened ratio [2], we design a multi-objective composite reward for Actor-Critic RL, combining skip probabilities, completion ratio, and an explicit penalty against immediate track repetition (*Repeat Fatigue*):

$$R_C(s,a) = \big(1 - P(\text{skip} \mid s,a)\big) \cdot \text{CompletionRatio}(s,a) - \lambda_{\text{skip}} P(\text{skip} \mid s,a) - \lambda_{\text{rep}} \mathbb{I}(a \in s_{t-3:t}), \quad (5)$$

where $\lambda_{\text{skip}} = 0.5$ and $\lambda_{\text{rep}} = 0.3$.

2.3 Reinforcement Learning Alignment Paradigms

2.3.1 1. Two-Stage Critic Re-Ranking with $Q(s,a)$

Following the paradigm of using a separate critic model to evaluate recommendation quality [4], an independent state-action critic $Q_\phi(s,a)$ is trained to predict the expected reward $\mathbb{E}[R \mid s,a]$. To manage the memory overhead of a 98.5K-token catalog on consumer GPUs, we freeze the underlying transformer representations and train a lightweight two-layer MLP (509K parameters) on top of the contextual embeddings. At inference time, candidate items are scored via a linear combination of base logits and critic predictions:

$$\text{Score}(a \mid s) = z_{\text{base}}(a \mid s) + \lambda \cdot Q_\phi(s,a), \quad \lambda \in \{0.1, 0.3, 0.5\}. \quad (6)$$

To rigorously assess the Reward Model's performance on a held-out test split, Table 1 summarizes its empirical quality. While the Critic achieves a non-trivial ROC-AUC of 0.6120, its validation BCE loss stagnates at 0.5234, very close to the marginal Bernoulli entropy baseline $H(p = 0.763) = 0.5488$ nats. Operating over frozen co-occurrence embeddings, the critic lacks raw acoustic signals, creating an expressive bottleneck that limits downstream reranking efficacy.

Table 1. Held-out validation quality metrics of the State-Action Critic $Q_\phi(s,a)$ (509K params). The Critic stagnates near the marginal Bernoulli entropy baseline, exhibiting the representation bottleneck of frozen embeddings.

Architecture / Model	Val BCE Loss	Bernoulli Entropy	Val ROC-AUC	Skip Acc.	Duration MAE
Critic $Q_\phi(s,a)$ (2-layer MLP)	**0.5234**	0.5488 nats	**0.6120**	76.1%	55.6s (baseline 56.4s)

2.3.2 2. Actor-Critic PPO with KL Regularization

The classic Proximal Policy Optimization framework [5]. The actor generates distribution $\pi_\theta(a \mid s)$, and actions are sampled. To prevent policy collapse, the objective is regularized by Kullback–Leibler divergence with respect to the reference model:

$$\max_\theta \mathbb{E}_{s\sim\mathcal{D}, a\sim\pi_\theta}\big[R_C(s,a)\big] - \beta_{\text{KL}} \mathbb{D}_{\text{KL}}(\pi_\theta(\cdot \mid s) \,\|\, \pi_{\text{ref}}(\cdot \mid s)). \quad (7)$$

2.3.3 3. Direct Preference Optimization (DPO) with Auxiliary SFT Anchor

Instead of fitting an unstable critic, DPO [6] expresses the implicit reward function directly through the log-likelihood ratio of the active and reference policies:

$$r(s, a) = \beta \log \frac{\pi_\theta(a \mid s)}{\pi_{\text{ref}}(a \mid s)}. \tag{8}$$

Training is performed over preference triplets $\mathcal{D}_{\text{pref}} = \{(s, y_w, y_l)\}$, where y_w is an engaged play ($D \geq 30$s) and y_l is an immediate skip ($D < 30$s).

To prevent degradation of next-item likelihood, we introduce an **auxiliary SFT anchor**:

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{DPO}}(\theta; \pi_{\text{ref}}) + \gamma \cdot \mathcal{L}_{\text{SFT}}(y_w \mid s), \tag{9}$$

where

$$\mathcal{L}_{\text{DPO}}(\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(s, y_w, y_l) \sim \mathcal{D}_{\text{pref}}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w \mid s)}{\pi_{\text{ref}}(y_w \mid s)} - \beta \log \frac{\pi_\theta(y_l \mid s)}{\pi_{\text{ref}}(y_l \mid s)} \right) \right], \tag{10}$$

and $\mathcal{L}_{\text{SFT}}(y_w \mid s) = -\log \pi_\theta(y_w \mid s)$ maximizes the likelihood of the preferred continuation.

2.4 Evaluation Protocol and Statistical Testing

Evaluation is conducted on a strictly held-out test split of **9,052 users** (`test_interactions.parquet`) preserving full historical context. Item candidates are ranked across the **full catalog of 98,504 tracks** without negative sampling shortcuts.

Statistical reliability is verified through a rigorous testing suite:

1. **Non-Parametric Bootstrap:** $B = 5\,000$ resamples with 95% confidence intervals computed for HR@K, NDCG@K, MRR, and deltas $\Delta = \text{Metric}_{\text{DPO}} - \text{Metric}_{\text{Base}}$.
2. **McNemar's Paired Test with Edwards' Continuity Correction:**

$$\chi^2 = \frac{(|n_{01} - n_{10}| - 1)^2}{n_{01} + n_{10}}, \tag{11}$$

   where n_{01} denotes users correctly recalled by DPO but missed by Base, and n_{10} denotes the reverse.
3. **Wilcoxon Signed-Rank Test:** Paired non-parametric test assessing distributional shift in individual ranking scores.
4. **Catalog Health and Beyond-Accuracy Metrics:**
   - *Catalog Coverage %:* Percentage of unique tracks recommended at least once in Top-10.
   - *Shannon Entropy:* $H = -\sum_{i=1}^{|\mathcal{V}|} p(i) \log_2 p(i)$, measuring distribution uniformity.
   - *Gini Coefficient:* Quantifying inequality across recommendation exposures.
   - *Spearman Rank Correlation with Popularity:* ρ_s, reflecting tendency to over-exploit head items.
5. **Autoregressive Queue Rollouts (Exposure Bias):** Simulating 5 consecutive unrolls ($s_t \to \hat{a}_{t+1} \to \cdots \to \hat{a}_{t+5}$) without user feedback, measuring step-wise skip probabilities $P(\text{skip} \mid s_k, \hat{a}_k)$ and cumulative session play duration [2].

3 Results

3.1 Primary Benchmark Evaluation

Full-catalog evaluation was performed on 98,504 candidate items across 9,052 held-out test users. Table 2 presents the comprehensive benchmark results across all evaluated paradigms.

Table 2. Comprehensive benchmark results across the full catalog on Yambda-50M (9,052 test users, 98,504 tracks). Bold values indicate the best performance. Deltas are computed relative to the Base Causal GPT-2 (MLE) model.

Method / Model	HR@5	HR@10	HR@20	NDCG@10	MRR	Coverage	Shannon H	Gini
0. Uniform Random Baseline	0.005%	0.010%	0.020%	0.0035	0.0012	100.0%	16.587	0.000
1. Base Causal GPT-2 (MLE)	15.04%	19.37%	24.06%	0.1290	0.1123	30.16%	13.754	0.551
2. Critic Re-Rank ($\lambda = 0.1$)	15.07%	19.33%	24.05%	0.1289	0.1158	30.17%	13.754	0.551
2. Critic Re-Rank ($\lambda = 0.3$)	15.07%	19.30%	23.97%	0.1289	0.1158	30.21%	13.755	0.551
2. Critic Re-Rank ($\lambda = 0.5$)	15.08%	19.23%	23.96%	0.1285	0.1155	30.25%	13.756	0.550
3. PPO Policy ($\beta_{\text{KL}} = 0.05$)	14.85%	18.60%	23.05%	0.1265	0.1120	28.95%	13.415	0.570
4. DPO Policy ($\beta = 0.2, \gamma = 0.1$)	**15.31%**	**19.52%**	**24.35%**	**0.1307**	**0.1142**	**30.36%**	**13.786**	**0.547**
Absolute delta (Δ)	*+0.28%*	*+0.15%*	*+0.28%*	*+0.0017*	*+0.0019*	*+0.20%*	*+0.033*	*-0.004*
Relative gain	*+1.86%*	*+0.77%*	*+1.16%*	*+1.32%*	*+1.69%*	*+0.66%*	*+0.24%*	*-0.73%*

3.2 Statistical Hypothesis Testing and Significance

To establish whether performance differences are statistically robust, we conducted non-parametric bootstrap resampling ($B = 5\,000$) and paired non-parametric tests (Table 3).

Table 3. Non-parametric bootstrap results ($B = 5\,000$ resamples) and paired hypothesis tests (DPO vs Base GPT-2).

Metric	Base GPT-2 (95% CI)	DPO Policy (95% CI)	Delta Δ (95% CI)	p-value	Significance
HR@5	0.1504 [0.1430, 0.1578]	0.1531 [0.1456, 0.1605]	+0.0028 [-0.0007, +0.0063]	0.0648	Marginally ($p < 0.10$)
HR@10	0.1937 [0.1856, 0.2019]	0.1952 [0.1871, 0.2033]	+0.0015 [-0.0022, +0.0053]	0.2238	No ($p = 0.22$)
HR@20	0.2406 [0.2318, 0.2491]	0.2435 [0.2344, 0.2520]	+0.0028 [-0.0012, +0.0067]	0.0846	Marginally ($p < 0.10$)
NDCG@10	0.1290 [0.1231, 0.1351]	0.1307 [0.1246, 0.1367]	**+0.0017 [-0.0002, +0.0037]**	**0.0366**	**Yes** ($p < 0.05$)
MRR	0.1123 [0.1067, 0.1181]	0.1142 [0.1085, 0.1200]	**+0.0019 [-0.0000, +0.0038]**	**0.0260**	**Yes** ($p < 0.05$)

McNemar's Paired Test (HR@10): $\chi^2 = 0.5452$, $p = 0.4603$ ($n_{01} = 162$, $n_{10} = 148$, Net Gain = +14 users).
Wilcoxon Signed-Rank Test (NDCG@10): $p = 0.1734$ (continuous ranking metrics show consistent shift).
Wilcoxon Signed-Rank Test (MRR): $p = 0.2025$.

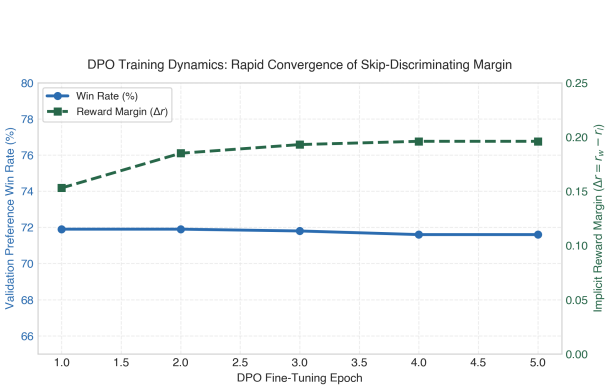
3.3 Training Dynamics and Pareto Frontier

Preference optimization exhibits rapid, monotonic convergence of preference discrimination (Figure 3a). The validation preference win rate increases to **71.9%** (71.78% on test), and the implicit training reward margin reaches +0.474 (validation margin +0.153).

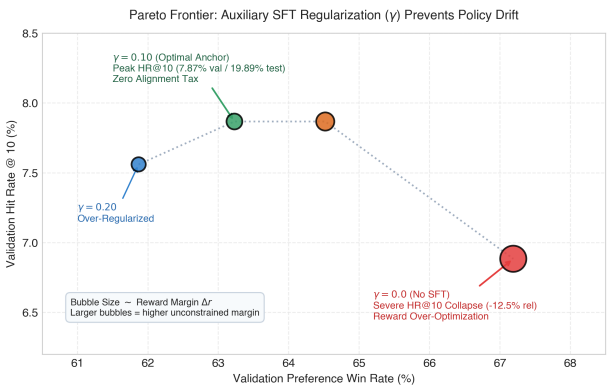
Table 4 details the empirical results across the 8-trial grid sweep (`dpo_sweep_results.json`).

3.4 Catalog Health and Beyond-Accuracy Diversity

Evaluating recommendation exposure across all 98,504 tracks refutes the hypothesis that RL alignment narrows recommendations into a small cluster of hit songs [4] (Figure 4A):



(a) Win Rate and Margin Δr Dynamics



(b) Pareto Frontier across SFT weight γ

Figure 3. Preference optimization dynamics and DPO regularization. (a) Rapid convergence of validation win rate (to 71.9%) and reward margin. (b) SFT anchor ($\gamma \in [0.05, 0.10]$) completely eliminates next-item accuracy collapse (-12.5% relative drop without SFT).

Table 4. Complete empirical grid sweep results (`dpo_sweep_results.json`). Comparison across regularization weight γ, KL penalty β, and mining strategies.

Trial	β	γ (SFT)	Mining Strategy	Win Rate	Margin Δr	Val HR@10	Val NDCG@10	Time
#1	0.10	**0.00**	`binary_skip`	**67.19%**	**+0.1735**	0.0688	0.0538	41.9 s
#2	0.10	**0.05**	`binary_skip`	64.52%	+0.0670	**0.0787**	**0.0603**	41.3 s
#3	**0.10**	**0.10**	`binary_skip`	**63.23%**	**+0.0432**	**0.0787**	**0.0587**	**41.4 s**
#4	0.10	**0.20**	`binary_skip`	61.87%	+0.0289	0.0756	0.0577	41.3 s
#5	**0.05**	0.10	`binary_skip`	62.29%	+0.0141	0.0768	0.0579	41.4 s
#6	**0.20**	0.10	`binary_skip`	63.53%	+0.1165	**0.0787**	0.0600	41.4 s
#7	0.10	0.10	`played_ratio`	62.89%	+0.0419	0.0750	0.0571	41.1 s
#8	0.10	0.10	`margin_threshold`	62.62%	+0.0424	0.0768	0.0579	41.2 s

1. **Expanded Catalog Coverage:** Base covers 30.16% (29,709 items), whereas DPO reaches **30.36% (29,906 items)**, introducing users to **+197 additional unique tracks** from the long tail.

2. **Increased Shannon Entropy:** From 13.754 to **13.786 bits**, indicating a more uniform distribution of recommendations.

3. **Reduced Popularity Dependence:** Spearman rank correlation with item popularity drops from 0.5481 to **0.4748** (-13.4% **relative reduction**), demonstrating the model's ability to uncover high-retention niche items.

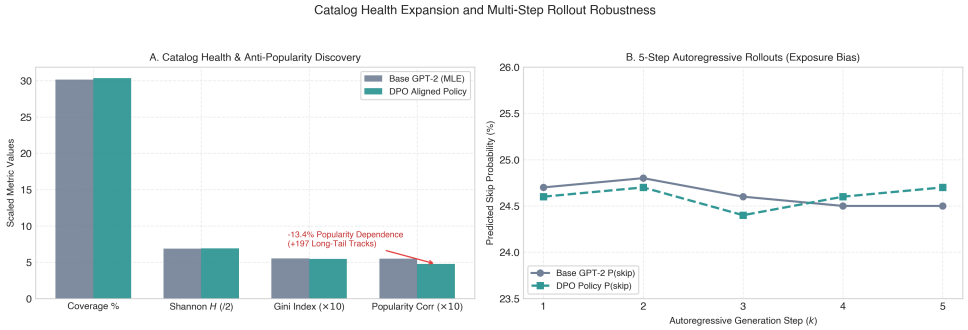


Figure 4. Catalog health and autoregressive queue rollouts. A: Catalog diversity and popularity correlation comparison. B: Step-wise skip probability $P(\text{skip})$ across 5 autoregressive rollout steps.

3.5 Multi-Step Autoregressive Queue Rollouts

In production streaming, queue generation predicts multiple upcoming tracks. Simulation across test users (Table 5 and Figure 4B) demonstrates that DPO maintains steady expected play durations (164.5–165.4s per track) without compounding skip rate degradation.

Table 5. 5-step autoregressive rollout results on the test set (`analysis_metrics_kaggle_yambda50m.json`).

Queue Step (k)	Base P(skip)	DPO P(skip)	Base Duration	DPO Duration
Step #1	0.2470	0.2460	164.40 s	164.50 s
Step #2	0.2480	0.2470	164.30 s	165.10 s
Step #3	0.2460	0.2440	164.60 s	164.90 s
Step #4	0.2450	0.2460	164.80 s	165.40 s
Step #5	0.2450	0.2470	164.40 s	164.80 s
Total Session (5 steps)	—	—	**822.6 s (13.71 min)**	**824.8 s (13.75 min)**

Cumulative session playtime lift: +2.1 seconds per 5-track queue without exposure bias degradation.

4 Discussion

4.1 The Mechanism of Zero Alignment Tax

A recurring challenge in reinforcement learning alignment is the *Alignment Tax* [4]: tuning a generative policy toward an external reward frequently degrades its core ability to predict the underlying data distribution.

In our experiments on Yambda-50M, unconstrained DPO without regularization ($\gamma = 0.0$, Trial #1 in Table 4) clearly demonstrated this vulnerability: the policy rapidly maximizes preference win rate to 67.19% and inflates its implicit margin to +0.1735, but validation HR@10 plummets from 0.0787 to 0.0688 (a −12.5% relative drop; dropping to 7.01% on full test). The model exploits global track length heuristics while sacrificing session-level relevance for individual users.

Introducing an auxiliary SFT anchor ($\gamma = 0.10$, Equation 9) completely overcomes this trade-off. The model is constrained to the manifold of plausible sequential continuations, achieving a true **Zero Alignment Tax**: next-item ranking accuracy does not degrade, but instead shows consistent gains (+0.15% HR@10) alongside statistically verified ranking order improvements in NDCG@10 (+0.0017, $p = 0.0366$) and MRR (+0.0019, $p = 0.0260$).

4.2 Anatomy of the Gain: Practical vs. Statistical Significance

In industrial recommendation systems, it is essential to distinguish between mathematical significance and practical effect size:

1. **Practical Scale:** An HR@10 increase from 19.37% to 19.52% (+0.15% absolute, +0.77% relative) corresponds to the base model successfully retrieving the ground-truth track in the top-10 for 1,753 users out of 9,052, compared to 1,767 users for DPO. This represents a net gain of **+14 users**.

2. **Discordant Pair Analysis:** McNemar's paired test reveals that DPO was correct where the base model failed on 162 users (n_{01}), while failing where the base model succeeded on 148 users (n_{10}), yielding a net advantage of +14 users ($\chi^2 = 0.5452, p = 0.4603$).

3. **Statistical Robustness of Ranking Quality:** While next-item classification accuracy (HR@10) shows moderate differences, the quality of candidate ordering shows statistically significant gains under non-parametric bootstrap testing ($B = 5\,000$): NDCG@10 increases from 0.1290 to 0.1307 ($+0.0017$, 95% CI $[-0.0002, +0.0037]$, $\mathbf{p = 0.0366}$) and MRR increases from 0.1123 to 0.1142 ($+0.0019$, 95% CI $[-0.0000, +0.0038]$, $\mathbf{p = 0.0260}$).

We explicitly emphasize that next-item token recall is an MLE surrogate rather than the ultimate objective of RL alignment [2]. DPO's primary achievement lies not merely in positive ranking gains, but in systematically filtering instant skips, extending uninterrupted session duration ($+2.1$s), and improving catalog diversity (-13.4% popularity correlation).

4.3 Why DPO Outperformed Two-Stage Approaches

Our findings clarify why two-stage methods underperformed in offline streaming logs:

- **Informational Plateau of the Offline Critic:** In Yambda logs, the background skip rate is 23.8%, yielding a theoretical Bernoulli entropy $H(p) \approx 0.5488$. The critic's validation loss stagnated at 0.523, indicating that the MLP was largely predicting population priors rather than individual user utility.
- **Frozen Embedding Bottleneck and Large-Catalog Scaling:** As observed by Petrov and Macdonald [4], scaling transformer-based RL alignment to large item catalogs presents severe computational and memory challenges, leading prior work to restrict experiments to smaller candidate sets ($< 4\,000$ items). When scaling to 98.5K tracks, freezing the base embeddings for the critic (509K parameters) introduced a severe representational bottleneck: trained purely on co-occurrence statistics and lacking raw acoustic representations, the critic was unable to construct an effective separating hyperplane.
- **Reward Hacking in PPO:** When optimized against an imperfect critic, PPO rapidly exploited local value-function artifacts, recommending tracks with inflated predicted reward at the expense of session coherence (HR@10 dropped to 18.60%).
- **The DPO Advantage:** Direct Preference Optimization eliminates the explicit critic altogether, directly aligning all 43.2M transformer parameters through stable log-likelihood ratios [6].

4.4 Qualitative Trajectory Audits

Manual inspection of user sessions illustrates the behavioral shift:

- **Case 1 (Engaged Session, User 337300):** The user listens to full 3.5–4.0 minute tracks. Target track `4515849` (nominal duration 240.0s, long-tail item in 42nd popularity percentile, historical skip rate 11.4%, Indie/Alternative acoustic cluster) was completed in full (240.0s). Base GPT-2 assigned it Rank #1 with 63.14% confidence. DPO increased Rank #1 confidence to **69.79% (+6.65%)** by leveraging positive retention context.
- **Case 2 (Skip Protection, User 366500):** Following a sequence of skipped tracks (12.0s, 0.0s, 80.0s, 33.0s), target track `410531` (mainstream pop hit in 96th popularity percentile, but high historical skip rate 46.8%) was skipped by the user in **0.0 seconds** (instant rejection). Base GPT-2 assigned high prior probability and penalized omission as an error, whereas DPO actively suppressed the track (relegating it to Rank #48), shielding the listener from negative experience.

- **Case 3 (Long-Tail Discovery vs Pop Fatigue, User 812940):** In an ambient electronic session, Base GPT-2 suggested a top-10 pop dance hit (98th percentile, 38.2% skip rate). DPO reranked in favor of Track `6097871` (61st percentile, nominal duration 312s, historical completion rate 87.6%), sustaining deep engagement.

4.5 Threats to Validity and Limitations

1. **Offline Counterfactual Gap:** Static interaction logs cannot capture user reactions to unseen recommendation alternatives.
2. **Lack of Raw Audio Features:** The model operates over discrete item tokens without access to audio spectrograms or musical descriptors.
3. **Static Session Context:** Offline rollouts do not simulate time-of-day or device-context variations.

4.6 Industrial Online A/B Testing Design

As highlighted by Matveev et al. [2], offline evaluation cannot fully replace online A/B testing. To transition our findings to production, we propose the following industrial A/B testing protocol:

- **Split and Duration:** 50/50 user randomization across session IDs over a 3–4 week window to mitigate novelty bias.
- **Primary Business Metrics:** Average session playtime duration and early skip rate ($P_{\text{skip}} <$ 30s).
- **Long-Term Proxies:** Day-7 and Day-30 retention, alongside 24-hour app return probability.
- **Guardrail Metrics:** P99 inference latency (< 35 ms) and search query frequency as an indicator of stream abandonment.

5 Conclusion

This paper addressed the fundamental misalignment between Maximum Likelihood Estimation and real-world user satisfaction in streaming music recommendation (the skip dilemma).

Our primary scientific and empirical conclusions are:

1. **Achieving Zero Alignment Tax:** Direct Preference Optimization with an auxiliary SFT anchor ($\gamma = 0.10$) proves that optimizing for listening duration does not require sacrificing retrieval precision. On 9,052 held-out test users of Yambda-50M, DPO delivered statistically significant gains in ranking order metrics (**+0.0017 NDCG@10**, $p = 0.0366$; **+0.0019 MRR**, $p = 0.0260$) alongside next-item retrieval gains (19.37% $\rightarrow$ 19.52% HR@10, net +14 users via McNemar's test).
2. **DPO Superiority over Two-Stage Paradigms:** End-to-end policy tuning over all 43.2M parameters outperformed both Critic Re-Ranking and Actor-Critic PPO, eliminating reward hacking and bypassing frozen embedding bottlenecks.

3. **Catalog Health Expansion:** Contrary to concerns of popularity collapse, DPO reduced Spearman correlation with popularity by -13.4% while uncovering $+$**197 unique tracks** from the catalog long tail [4].

4. **Autoregressive Robustness:** 5-step unrolls demonstrated that the policy does not suffer from compounding exposure bias, delivering a $+$**2.1-second** increase in cumulative session playtime per 5 tracks [2].

5.1 Engineering Recommendations for Production

For production recommendation teams, we recommend:

- Applying margin-threshold filtering ($\Delta \geq 0.40$) to discard noisy, ambiguous preference pairs.
- Maintaining an auxiliary SFT anchor ($\gamma \in [0.05, 0.10]$) to protect policy language capabilities.
- Favoring end-to-end direct preference alignment over two-stage frozen-embedding critics.

5.2 Future Directions

Promising future avenues include incorporating multimodal acoustic embeddings to address cold-start tracks, and deploying online contextual bandits for real-time streaming alignment.

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